Experiment No. 7

Aim: To develop an **Activity Diagram** for a selected software project using an open-source UML diagramming tool.

Objectives

1. To understand the concept of Activity Diagrams.
2. To identify activities, decisions, and workflows in a software system.
3. To represent the sequence of activities graphically.
4. To develop an Activity Diagram using an open-source tool.

Software Required

- Web browser
- SmartDraw
- SmartDraw account, if required

Theory

An **Activity Diagram** is a UML behavioral diagram used to represent the **workflow of a system or business process**. It shows the sequence of activities, decisions, alternative paths, and the starting and ending points of a process.

Main Elements of an Activity Diagram

Element	Description
Initial Node	Indicates the starting point of the process.
Activity/Action	Represents a task or operation performed by the system or user.
Decision Node	Represents a condition that creates alternative paths.
Control Flow	Arrow showing the sequence between activities.
Fork	Divides one flow into multiple parallel activities.
Join	Combines multiple parallel flows into one flow.
Final Node	Indicates the end of the process.

Case Study: Write case study name and draw flow of diagram

Procedure: Write procedure steps for activity diagram using smartdraw tool

Conclusion

The experiment demonstrated how an **Activity Diagram** can be used to represent the workflow of a software project. The activities, decisions, control flows, and start/end points of the Library Management System were successfully modeled using an open-source tool.